

# One Record per Click: Serving Capacity Selects the Training Target in Delayed-Feedback Conversion Prediction

Anonymous Author<sup>1</sup>

Anonymous Institution, Anonymous City, Country  
anonymous@example.org

**Abstract.** Many production conversion-rate (CVR) pipelines write exactly one training record per click: it is emitted after a fixed wait  $\Delta$ , carries the label  $1[\text{conversion} \leq \text{click} + \Delta]$ , and is never revisited. We show that this contract reduces the choice of training target to a one-dimensional design problem — how much of the target is a realised hard label and how much a model-produced soft one — and that the served model’s capacity selects where on that axis to sit. Instrumenting the feature path of fourteen published methods over a full 720-hour replay of the Criteo Delayed Feedback Dataset (Criteo-DFD), we find that nine emit more than one record per click, including the VANILLA baseline the field normalises against, so their reported gains are not attainable under the contract. Sweeping target hardness over the admissible remainder against a five-rung serving-capacity ladder (26 K–9.6 M parameters) yields two ordered crossovers. At the smallest rung a purely soft target that uses no realised label at all is the best arm, beating even the non-causal Oracle by 0.0029 area under the ROC curve (AUC); by 46 K the hybrid targets of prior work overtake it; and beyond  $10^5$  parameters the Oracle’s true label is best, with the soft target now costing 0.0084. A published baseline’s prescribed auxiliary learning rate, meanwhile, is worth 0.0022 AUC to it — roughly seven times the gap between the methods it is used to compare.

**Keywords:** Conversion rate prediction · Delayed feedback · Knowledge distillation · Online advertising.

## 1 Introduction

Conversion-rate (CVR) prediction decides what an advertising system shows and what it charges, and it is trained continuously on a stream. Its central difficulty is that the label arrives late: a click is observed immediately, but the conversion it may cause can follow minutes, days or weeks afterwards [1], so a model trained on the labels observed so far is trained on a censored, downward-biased target. The literature treats this as an estimation problem, and a decade of work has produced importance weighting [14, 13], label correction [2, 11], duplicated replay streams [6, 5, 3], multi-window ensembles [7, 9, 15] and influence-function retrofits [4]. Almost all share an unstated systems assumption: that a click may be written into the training stream more than once — re-emitted

when its conversion lands, replayed when its window closes, or relabelled at several horizons.

Many production pipelines cannot do this. A common contract is that *a click produces exactly one training example, emitted after a fixed wait  $\Delta$ , carrying the label  $1[\text{conv\_ts} \leq \text{click\_ts} + \Delta]$ , and never revisited* — so if the conversion has not arrived by then, the example is negative permanently. The reasons are mundane and therefore durable: amending a materialised example later means mutable training data, duplicate-key handling, and a reproducibility story that breaks on every backfill, whereas freezing the label at  $\Delta$  keeps the dataset append-only and the pipeline replayable. The contract is not our invention. It is the literal reading of the field’s own reference baseline, which “trains each click *once* at base observation age *o*” [8]. What we show is that the implementations do not honour it — including, as Sect. 5.2 establishes, that baseline’s own.

**One record, one question.** The contract is restrictive enough to be clarifying. If a click yields a single record carrying a single target, the only remaining freedom is *what that target says*. At one extreme it is a realised outcome: a hard 0/1 draw, unbiased but high-variance, and correct only for conversions fast enough to have landed. At the other it is a model’s estimate: a soft probability, lower-variance and defined for every record, but capped by the estimator that produced it. Every admissible method is a point between these two. FSIW [14] reweights the hard label, ULC [11] blends it with a learned correction, and TWICE [8] routes the residual to a separate head. We call this quantity the *hardness* of the target, and we show that the best choice along it is not a property of the dataset but of the *served model’s capacity* — the axis a latency-bound ranker is actually constrained on, and the one the delayed-feedback literature holds fixed.

### Contributions.

1. **A measurement, not an assertion.** We instrument the feature path of fourteen methods and count training records per click over a full replay (Sect. 5.2). Nine exceed one. Two measurement traps — traffic variation and window length — both *understate* violations; we control for both.
2. **A reformulation.** Under the contract the admissible targets form a single hardness axis spanned by two label pipelines (Sect. 3), and a purely soft target satisfies the frozen-label obligation *vacuously*, its record carrying no label to freeze (Proposition 1).
3. **The main result.** Optimal hardness increases with serving capacity, through two ordered and significant crossovers (Sect. 5.4): the soft target cedes to the hybrid between 26 and 46 K parameters, and the hybrid cedes to the Oracle’s true label near  $10^5$ . Below that second crossing the non-causal Oracle is *not* an upper bound.
4. **A reproducibility finding.** Baseline auxiliary learning rates do not transfer: one baseline’s prescribed rate costs it 0.0022 AUC while the same change *harms* another (Sect. 5.3), so the admissible ordering cannot be read without per-method tuning.

## 2 Related Work

We organise prior work by whether it can be expressed as one record per click, because that is what the constraint decides. Sect. 5.2 gives measured counts; here we give the mechanism.

**Methods that need more than one record.** The fake-negative family writes the click as a negative and re-emits it as a positive when the conversion arrives [6]; ES-DFM samples an elapsed-time cut-off and duplicates the delayed positive [13], DEFER replays real negatives once their window closes [5], DDFM runs both streams together [3], and DEFUSE adds a label correction on top of the duplicated stream [2]. Multi-window methods relabel the same click at several horizons: FTP trains  $K$  maturity-gated tasks and a policy over them [7], MISS duplicates each delayed positive into every head’s stream [9], and PI interpolates two fixed-window predictors [15]. IF-DFM applies an influence-function update when a label reverses [4]. In each case the extra visit *is* the mechanism, not an implementation detail that could be removed.

**Single-record methods, and where they sit on the axis.** A smaller group reweights or relabels the one fresh record, and these are what the constraint leaves standing. FSIW keeps the realised label and importance-weights the loss [14]; ULC adds a learned correction to it, giving a target that is part observation and part estimate [11]; TWICE keeps its CVR tower on a single fresh record and routes conversion-clock events to a separate delay head [8]. All three retain the record’s own realised label; none removes it. The end of the axis at which the target is a model estimate alone is, to our knowledge, unmeasured in this literature, and Sect. 5.4 shows it is the best admissible choice at small serving capacity.

**Distillation with privileged information.** Training-time-only signals have been distilled into serving models before, notably privileged features at Taobao [12], and soft labels carry lower target variance than a hard Bernoulli label [10] — the mechanism we invoke in Sect. 5.4. The privilege here is neither a feature nor scale but *time*: the estimator is allowed to wait.

## 3 Problem Formulation

### 3.1 Notation and two clocks

**Notation.** For click  $i$ ,  $x_i$  are its features and  $C_i$  its click time; if a conversion follows,  $V_i$  is its arrival time and  $D_i = V_i - C_i$  the delay, with  $D_i = \infty$  otherwise. The business *attribution window*  $v$  fixes the served target  $Y_i^{(v)} = 1[D_i \leq v]$ , knowable only at  $C_i + v$ . The pipeline emits click  $i$ ’s record after a wait  $\Delta$ ; at the shortest usable wait, the *base observation age*  $o \ll v$ , the record can carry  $Y_i^{(o)} = 1[D_i \leq o]$ . We write  $p_v(x) = P(Y^{(v)}=1 \mid x)$  for the estimand and  $\hat{p}_v$  for a model of it,  $w(x) = P(Y^{(v)}=1 \mid Y^{(o)}=0, x)$  for the residual correction shared by FSIW, ULC and ES-DFM, and  $\tilde{y}_i$  for the target actually written on the record. Following the long-horizon formulation of [8], clicks are observed on a *click clock* and conversions arrive on a separate *conversion clock*: training must predict  $Y^{(v)}$  from a record written before  $Y^{(v)}$  is known.

### 3.2 The single-visit constraint

Let  $\Delta$  be the pipeline’s wait. The contract of Sect. 1 imposes two obligations:

- R1 (single visit)** click  $i$  contributes exactly one training record, ever;
- R2 (frozen label)** that record’s label is  $1[D_i \leq \Delta]$  for the  $\Delta$  at which it was emitted; no later-arriving conversion may add or change supervision.

A method may fail either independently: duplication fails R1, and reading  $Y^{(v)}$  on a record released before  $C_i + v$  fails R2.

### 3.3 Two label pipelines

Because a record is written once and  $\Delta$  is the only free quantity, the constraint admits exactly two useful pipelines over the same clicks:

$$\mathcal{D}_{\text{fresh}} = \{(x_i, Y_i^{(o)})\}, \quad \Delta = o \ll v \quad \text{timely, biased low,} \quad (1)$$

$$\mathcal{D}_{\text{mat}} = \{(x_i, Y_i^{(v)})\}, \quad \Delta = v \quad \text{unbiased, stale by } v. \quad (2)$$

For  $\mathcal{D}_{\text{fresh}}$ ,  $\mathbb{E}[Y^{(o)} \mid x] = p_v(x) F(o \mid x) < p_v(x)$ , so it is biased low by exactly the fraction of converters not yet observed. On Criteo-DFD at  $o = 1$  h that fraction is 0.77, 0.57 and 0.47 at  $v = 1, 7$  and 30 d: censoring deepens as the window lengthens, yet even at  $v = 30$  d the fresh label still carries almost half the target’s positives, which makes this corpus a conservative setting for any method that discards it.  $\mathcal{D}_{\text{mat}}$  carries the true target but its most recent row is  $v$  old.

R1 forbids training the served model on both. What remains is to write one record at  $\Delta = o$  and use  $\mathcal{D}_{\text{mat}}$  to shape *what its target says*.

### 3.4 The target-hardness axis

Every admissible target is a function of the record’s own realised label  $Y_i^{(o)}$  and of quantities estimated on  $\mathcal{D}_{\text{mat}}$ . Writing the general single-record form

$$\tilde{y}_i = Y_i^{(o)} \alpha(x_i) + (1 - Y_i^{(o)}) \beta(x_i), \quad (3)$$

with  $\alpha, \beta$  fitted on  $\mathcal{D}_{\text{mat}}$  and frozen at emission, gives a one-dimensional family ordered by how much the target depends on the realised label. We name three points on it, and they are the arms we compare:

- hard** the target *is* a realised label:  $\alpha=1, \beta=0$  gives  $\tilde{y} = Y^{(o)}$  (FRESH). Two arms supply a realised outcome from outside this  $\Delta=o$  family and we group them here too: WAIT emits  $\tilde{y} = Y^{(v)}$  at  $\Delta = v$ , and the non-causal ORACLE uses  $Y^{(v)}$  at  $\Delta = o$ , where it is unknowable;
- hybrid** realised label plus a learned correction:  $\alpha=1, \beta=w$  gives  $\tilde{y} = Y^{(o)} + (1 - Y^{(o)})w(x)$ , which is exactly ULC [11];
- soft** a model estimate alone:  $\alpha=\beta=\hat{p}_v$  gives  $\tilde{y} = \hat{p}_v(x)$ , which uses no realised label at all.

FSIW [14] sits at the hard end by a different route, leaving  $\tilde{y} = Y^{(o)}$  and reweighting the loss instead of the target.

**Proposition 1.** *Let  $\tilde{y}_i$  follow (3) with  $\alpha, \beta$  fitted on  $\mathcal{D}_{\text{mat}}$  and frozen at emission time  $C_i + o$ . Then (i)  $\tilde{y}_i$  is computable at  $\Delta = o$  and the record satisfies R1 and R2; and (ii) if  $\alpha \equiv \beta$ , the record satisfies R2 vacuously: its target is a function of  $x_i$  alone, so it carries no realised label that a later-arriving conversion could revise.*

*Proof.* (i) The record is written once, at  $\Delta = o$ , so R1 holds. Its target is a deterministic function of  $(x_i, Y_i^{(o)})$  and of parameters frozen at emission;  $Y_i^{(o)}$  is knowable at  $C_i + o$  and is never revised, so R2 holds. (ii) If  $\alpha \equiv \beta$  then (3) collapses to  $\tilde{y}_i = \alpha(x_i)$ , which does not depend on  $Y_i^{(o)}$ . R2 constrains the realised label carried by the record, and there is none.  $\square$

Proposition 1(ii) is the structural reason to care about the soft end beyond accuracy: a record with no realised label needs no label join on the timely path. We return to this in Sect. 6.

## 4 Instantiating the Axis

Realising the hybrid and soft points of Sect. 3.4 requires an estimator of  $w$  and of  $p_v$  that is itself admissible. We fit one model on the matured pipeline and read both from it, so that a comparison across the axis changes the target and nothing else.

The estimator trains only on clicks whose window has closed,  $C_i + v \leq T_n$  at cutoff  $T_n$ , under the true label  $Y^{(v)}$ . It is causal by construction — every training row’s outcome was knowable at  $T_n$  — and is itself R1/R2-compliant at  $\Delta = v$ . It carries two read-outs on one trunk:

$$\hat{p}_v(x) = P(Y^{(v)} = 1 \mid x), \quad w^T(x) = P(Y^{(v)} = 1 \mid Y^{(o)} = 0, x). \quad (4)$$

The second is the estimand the field shares — it is ULC’s  $w$ , ES-DFM’s delayed-positive rate and DDFM’s  $z_1$ . Substituting (4) into (3) gives the two admissible arms we sweep: the hybrid  $\tilde{y} = Y^{(o)} + (1 - Y^{(o)}) w^T(x)$ , and the soft  $\tilde{y} = \hat{p}_v(x)$ . Because the estimator is training-only it is unbound by the serving budget; that asymmetry is what buys the right to wait. Varying it at a fixed served model (S2) shows that carrying the second read-out is free, so one estimator can supply both arms, and that enlarging it from 9.5 M to 57 M parameters does not help: Criteo-DFD’s nine categorical fields hold 135,151 distinct values against a table provisioned at 131,072 rows *per field*, so the estimator is already at the corpus’s resolution and its saturation point coincides with its optimum.

Two implementation points matter and are easy to get wrong; both cost us a run before they were fixed. First, the estimator’s learning rate must be *decoupled* from the served model’s: sharing it makes a per-arm sweep tune two models with one scalar, and in our harness that made FSIW degenerate to fresh-only behaviour (log-loss 0.514, the fresh-only signature) while the sweep railed at the top of the grid, selecting for auxiliary convergence rather than CVR fit. Second, both read-outs must be withheld until the estimator has seen a minimum number of matured rows — the hybrid falling

back to  $\tilde{y} = Y^{(o)}$  and the soft arm likewise — because a zero-initialised head asserts 0.5 on every row, and a constant target yields a loss of  $\ln 2$  at every learning rate. Without the guard the soft arm’s rate is selected on a degenerate objective; Sect. 7 records the one window where this bites even with the guard in place.

## 5 Experiments

### 5.1 Setup

**Dataset and protocol.** Criteo-DFD [1], 60 days of clicks with conversion timestamps. We follow the streaming replay of [8]: pre-train on days  $[0, 30)$ , then stream  $[30, 60)$  with hourly cut-offs; at each cut-off a model consumes the records released in that hour and then predicts every click of the next hour exactly once, scored retrospectively against  $Y^{(v)}$ . Base observation age  $o = 1$  h; we report  $v = 7$  d in the main text and  $v = 1$  d as a second window.

**Evaluation metrics.** Following [8], every metric is computed per next-hour cohort and sample-count weighted over the streaming period. We report area under the ROC curve (AUC,  $\uparrow$ ); average precision as PR-AUC ( $\uparrow$ ); normalised entropy  $NE = LL/H(\bar{y})$  ( $\downarrow$ ), the mean binary cross-entropy divided by the entropy  $H(\bar{y}) = -\bar{y} \log \bar{y} - (1-\bar{y}) \log(1-\bar{y})$  of the evaluation base rate  $\bar{y}$ , which makes the scale comparable across windows whose base rates differ; and calibration  $Cal = \sum_i \hat{p}_i / \sum_i y_i$ , the predicted-over-actual ratio, for which 1 is perfect and  $< 1$  under-predicts. For relative improvement we use TWICE’s Eq. (23) with the anchor  $A$  made explicit,

$$RI^A(M) = \frac{M - M_A}{M_{\text{ORACLE}} - M_A}, \quad (5)$$

the fraction of the  $A$ -to-Oracle gap a method recovers, computed per seed from matched  $A$  and Oracle runs and then averaged; NE reductions give positive RI because both differences are negative. TWICE sets  $A = \text{VANILLA}$ . We report  $A = \text{WAIT}$  and  $A = \text{FRESH}$  instead, the two *admissible* endpoints, for reasons Sect. 5.4 sets out; each is run at every rung with its own selected learning rate, so every RI is normalised by the headroom at its own capacity. All numbers are means over five seeds and every comparison is a paired  $t$ -test over matched seeds ( $*p < 0.05$ ,  $**p < 0.01$ ).

**Compared methods.** Table 2 names every arm and its point on the axis. REWEIGHT is FSIW [14], CORRECT is ULC [11], WAIT trains on  $\mathcal{D}_{\text{mat}}$  only, and DISTIL and SOFT use the estimator of Sect. 4. VANILLA, ES-DFM [13], DEFUSE [2] and DDFM [3] are shown for reference but are inadmissible (Sect. 5.2). The non-causal Oracle labels the fresh record with  $Y^{(v)}$ .

**Implementation details.** Minibatch 8192, AdamW, and a learning rate selected *per arm and per capacity* from  $\{10^{-6}, 2.5 \times 10^{-6}, 10^{-5}, 5 \times 10^{-5}, 2 \times 10^{-4}, 5 \times 10^{-4}, 10^{-3}\}$  on a held-out chronological slice of the pre-training period. The served model is swept over five rungs S0–S4 varying the categorical hash-table size ( $36 \rightarrow 131,072$  rows,  $26 \text{ K} \rightarrow 9.6 \text{ M}$  parameters) at a fixed embedding dimension of 8, so the ladder is a pure serving-capacity axis. The matured-data estimator is held at 9.5 M parameters throughout. At S4 the served model and the estimator are within 1.3% of the same parameter count, which is the cleanest available control for capacity transfer. Auxiliary learning rates are swept per method for the reason established in Sect. 5.3.

**Table 1.** Training records per click, measured over a full 720-hour replay at  $v = 7$  d. Counts are per *distinct click touched*, so traffic variation cannot inflate them, and the window covers every method’s longest revisit lag. <sup>†</sup> TWICE’s served tower is single-visit; its delay head reads the conversion clock

| method              | records/click | max | R1 | R2             |
|---------------------|---------------|-----|----|----------------|
| <i>admissible</i>   |               |     |    |                |
| Fresh               | 1.00          | 1   | ✓  | ✓              |
| Wait / teacher      | 1.00          | 1   | ✓  | ✓              |
| Reweight            | 1.00          | 1   | ✓  | ×              |
| Correct             | 1.00          | 1   | ✓  | ×              |
| Twice               | 1.00          | 1   | ✓  | ✓ <sup>†</sup> |
| <i>inadmissible</i> |               |     |    |                |
| Vanilla             | 1.08          | 2   | ×  | ×              |
| DFM                 | 1.34          | 8   | ×  | ×              |
| ES-DFM              | 1.08          | 2   | ×  | ×              |
| DEFUSE              | 1.08          | 2   | ×  | ×              |
| DDFM                | 1.89          | 2   | ×  | ×              |
| PI                  | 2.00          | 2   | ×  | ×              |
| MISS                | 2.05          | 3   | ×  | ×              |
| IF-DFM              | 2.90          | 99  | ×  | ×              |
| FTP                 | 3.00          | 3   | ×  | ×              |

## 5.2 RQ1: which methods can be run at all?

Table 1 reports the measurement. We instrument the feature path of every arm and count training records over a full 720-hour replay, charging only gradient-carrying forwards so that merely *scoring* a historical click is not counted. Two traps had to be controlled, and ignoring either understates violations.<sup>1</sup>

Nine of fourteen methods exceed one record per click. Two counts are exact integers and confirm the instrument: PI emits 2.00, one per fixed-window head, and FTP 3.00, one per task plus its prophet. IF-DFM touches one click up to 99 times. Five arms stay at one record, two of them the uncorrected FRESH and WAIT: among methods that correct for delay, only FSIW, ULC and TWICE keep the served model single-visit.

**The field’s anchor is inadmissible.** VANILLA — “each click trained once at base observation age  $o$ ” — measures 1.08 records per click, because matching any published VANILLA row requires also emitting the late conversion as a positive.<sup>2</sup> Relative

<sup>1</sup> *Traffic*: per-hour ratios conflate duplication with diurnal variation, since a maturity-gated stream reads clicks from  $v$  ago when the rate differed; PI scores a spurious 12.2 that way. We divide by *distinct clicks touched*. *Window*: over 48 h PI measures 1.00, its two visits being 10.5 days apart. We run the full stream and score only clicks whose entire revisit horizon fits inside it.

<sup>2</sup> At the  $v = 30$  d setting the published rows use, the fresh record alone predicts 0.107 against a 0.227 base rate and reproduces no published VANILLA log-loss; the reading that does is fresh label plus a positive for conversions arriving after  $o$ .

improvement reported against it is therefore normalised by a baseline a single-visit pipeline cannot run.

**On auxiliaries.** FSIW, ULC and TWICE emit one record for the served model but fit a *separate* model on matured data, so they satisfy R1 and sit at the boundary of R2. We read the constraint as governing the served model’s pipeline, under which they comply; the alternative reading — that no later-arriving conversion may inform anything — leaves only WAIT standing and declares the problem unsolvable rather than selecting a method. Every arm we compare sits at exactly this level and no lower.

### 5.3 RQ2: is the admissible ordering readable?

Before comparing targets we must establish that the comparison is about targets. Each single-record baseline prescribes a fixed learning rate for its auxiliary, and these are not interchangeable. Holding everything else fixed and moving CORRECT’s auxiliary from its prescribed  $10^{-3}$  to  $5 \times 10^{-4}$  raises it by  $+0.0005^*$ ,  $+0.0004^{**}$ ,  $+0.0002^*$ ,  $+0.0002^{**}$  and  $+0.0022^{**}$  across S0–S4. The same change applied to REWEIGHT harms it, by  $-0.0005^{**}$ ,  $-0.0004^{**}$  and  $-0.0036^{**}$  at S0, S2 and S3.

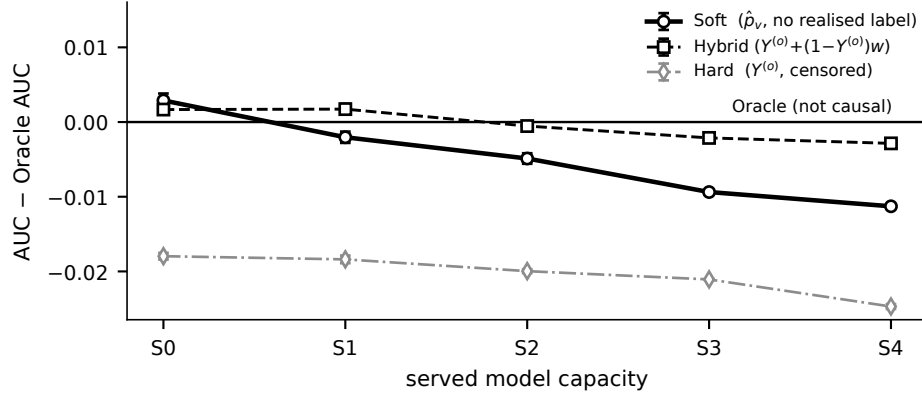
Two consequences follow. First, the effect is large relative to what it is used to measure: at S4 the tuning alone moves CORRECT by 0.0022, roughly seven times the 0.0003 that separates it from DISTIL once both are tuned. A comparison at prescribed rates would have reported a 0.0025 method gap that is 88% tuning. Second, because the same change helps one baseline and hurts another, the bias is not systematic and cannot be corrected globally. We therefore sweep auxiliary rates per method and report tuned values throughout. This bounds how finely published orderings in this literature can be read.

### 5.4 RQ3: optimal target hardness tracks serving capacity

Table 2 and Fig. 1 are the main result, and the table is read down the columns rather than across the rows: the best admissible target changes with capacity.

**First crossover: soft cedes to hybrid between 26 and 46 K parameters.** At S0 the soft target is the best admissible arm, ahead of the hybrid by  $+0.0012^*$ . By S1 the ordering has reversed, and it stays reversed and widens monotonically:  $-0.0038^{**}$ ,  $-0.0043^{**}$ ,  $-0.0072^{**}$  and  $-0.0084^{**}$  at S1–S4. The soft arm’s own ladder is nearly flat ( $0.8285 \rightarrow 0.8352$  across the five rungs) where the hybrid’s climbs ( $0.8273 \rightarrow 0.8437$ ), which is the mechanism: a purely soft target is a *function of  $x$* , so two records with identical features receive identical supervision, and a larger model can only fit that function more exactly. The realised label carries per-example information that no function of  $x$  reproduces. A small model cannot exploit that information and is hurt by its Bernoulli variance [10]; a large one can, and discarding it becomes pure loss.

**Second crossover: the Oracle is not an upper bound below  $10^5$  parameters.** The hybrid beats the non-causal Oracle at S0 and S1, by  $+0.0017^{**}$  at both, and loses from S2 on ( $-0.0005^*$ ,  $-0.0021^{**}$ ,  $-0.0029^{**}$ ). The soft target beats it more decisively still at S0, by  $+0.0029^{**}$ . Below roughly  $10^5$  parameters a corrected or estimated target is therefore *better supervision than the true label*, for the same variance reason. This



**Fig. 1.** Both crossovers, plotted as a margin against the non-causal Oracle so that the Oracle is the zero line; bars are 95% intervals over five paired seeds. The soft and hybrid targets cross *each other* between S0 and S1, and both cross *zero* between S1 and S2: below roughly  $10^5$  served parameters an estimated target beats the true one

matters for how the literature evaluates: an assessment conducted only at large capacity reports a delayed-feedback penalty that a deployable model does not fully pay.

**Within the hybrid class, the source of the correction barely matters.** DISTIL and CORRECT differ by  $-0.0007$  to  $+0.0004$  across the ladder, reaching  $p < 0.05$  only at S3 and S4 and never exceeding 0.0003 there. The two correctors hold the same parameter count to within 0.13% (9,444,801 against 9,456,897) and share a fit schedule, so this parity is measured with capacity and tuning both controlled. The choice of hybrid estimator is not where the value lies; the choice of *hardness class* is.

**Compliance is almost free on this corpus.** The best admissible arm leads the best inadmissible one at every rung on PR-AUC ( $+0.0010$  to  $+0.0042$ , all  $p < 0.01$ ) and on AUC ( $+0.0001$  to  $+0.0014$ ). The single exception is NE at S4, where DDFM leads by 0.0006 ( $p = 0.019$ ); at S0–S3 the admissible arm leads on NE too. Duplication therefore buys almost nothing here, and what little it buys appears only in the likelihood at the largest capacity.

**Calibration favours the soft end, and confirms the formulation.** FRESH predicts 0.579 of the actual conversion rate, within 1% of the censoring factor  $F(o) = 0.574$  that Sect. 3 predicts for  $v = 7$  d — the bias of  $\mathcal{D}_{\text{fresh}}$  measured directly. Among corrected arms the soft target is the best calibrated at every rung (1.007 at S0 against 1.031 for both hybrids and for the Oracle itself), since it distils an already-calibrated probability instead of correcting a realised label. Where  $\hat{p}_v$  is priced rather than merely ranked, that favours the soft end beyond what AUC shows.

**What RI measures, and what it does not.** Relative improvement is the field’s usual way of putting a small absolute gain in context, so it is worth establishing what it can carry here. Two properties matter, and the first is exact rather than empirical.

Within a rung,  $\text{RI}^A(M) = (M - M_A)/(M_{\text{ORACLE}} - M_A)$  is *affine in M with positive slope*, since  $M_{\text{ORACLE}} > M_A$  for every anchor we consider. RI is therefore a

**Table 2.** Accuracy and calibration against serving capacity,  $v = 7$  d, five seeds, arms grouped by their point on the hardness axis of Sect. 3.4. Leading zeros are omitted. Bold marks the best *admissible* arm per column; the Oracle is not causal and the inadmissible block cannot be run at all, so neither competes for it, and calibration has no “best” so none is marked. On AUC the bold entry moves from the soft target at S0 to the hybrid from S1 on

| method                                             | NE ↓         |              |              |              |              | AUC ↑        |              |              |              |              | PR-AUC ↑     |              |              |              |              | Cal. ( $\rightarrow 1$ ) |       |       |       |      |
|----------------------------------------------------|--------------|--------------|--------------|--------------|--------------|--------------|--------------|--------------|--------------|--------------|--------------|--------------|--------------|--------------|--------------|--------------------------|-------|-------|-------|------|
|                                                    | S0           | S1           | S2           | S3           | S4           | S0           | S1           | S2           | S3           | S4           | S0           | S1           | S2           | S3           | S4           | S0                       | S1    | S2    | S3    | S4   |
| <i>hard: target is a realised label</i>            |              |              |              |              |              |              |              |              |              |              |              |              |              |              |              |                          |       |       |       |      |
| FRESH                                              | .8704        | .8659        | .8599        | .8551        | .8532        | .8077        | .8127        | .8182        | .8227        | .8218        | .5472        | .5509        | .5555        | .5599        | .5581        | .579                     | .579  | .579  | .579  | .58  |
| REWEIGHT                                           | .7645        | .7572        | .7480        | .7395        | .7376        | .8218        | .8269        | .8334        | .8388        | .8396        | .5656        | .5717        | .5783        | .5843        | .5844        | 1.026                    | 1.025 | 1.025 | 1.022 | 1.02 |
| WAIT                                               | .7821        | .7765        | .7684        | .7580        | .7522        | .8105        | .8156        | .8212        | .8286        | .8321        | .5488        | .5526        | .5589        | .5659        | .5702        | 1.003                    | 1.001 | 1.000 | 1.001 | 1.00 |
| <i>hybrid: realised label + learned correction</i> |              |              |              |              |              |              |              |              |              |              |              |              |              |              |              |                          |       |       |       |      |
| TWICE                                              | .7831        | .7725        | .7646        | .7582        | .7551        | .8120        | .8217        | .8260        | .8305        | .8327        | .5505        | .5590        | .5641        | .5681        | .5699        | .949                     | .986  | .969  | .971  | .96  |
| CORRECT                                            | <b>.7549</b> | .7486        | <b>.7413</b> | .7348        | .7313        | .8281        | .8325        | .8374        | .8414        | .8434        | <b>.5744</b> | .5789        | <b>.5843</b> | .5891        | .5918        | 1.027                    | 1.027 | 1.028 | 1.026 | 1.02 |
| DISTIL                                             | .7563        | <b>.7482</b> | .7415        | <b>.7347</b> | <b>.7311</b> | .8273        | <b>.8328</b> | <b>.8377</b> | <b>.8416</b> | <b>.8437</b> | .5732        | <b>.5793</b> | .5841        | <b>.5892</b> | <b>.5920</b> | 1.031                    | 1.031 | 1.031 | 1.030 | 1.02 |
| <i>soft: model estimate alone</i>                  |              |              |              |              |              |              |              |              |              |              |              |              |              |              |              |                          |       |       |       |      |
| SOFT                                               | .7571        | .7566        | .7504        | .7515        | .7495        | <b>.8285</b> | .8291        | .8333        | .8344        | .8352        | .5675        | .5682        | .5727        | .5732        | .5743        | 1.007                    | 1.009 | 1.008 | 1.010 | 1.00 |
| <i>inadmissible: &gt; 1 record per click</i>       |              |              |              |              |              |              |              |              |              |              |              |              |              |              |              |                          |       |       |       |      |
| VANILLA                                            | .7687        | .7610        | .7519        | .7430        | .7414        | .8222        | .8280        | .8347        | .8403        | .8412        | .5644        | .5700        | .5766        | .5827        | .5829        | .910                     | .909  | .907  | .904  | .90  |
| ES-DFM                                             | .7713        | .7554        | .7457        | .7406        | .7345        | .8166        | .8288        | .8357        | .8383        | .8421        | .5593        | .5744        | .5813        | .5827        | .5876        | 1.028                    | 1.026 | 1.023 | 1.020 | 1.01 |
| DEFUSE                                             | .7704        | .7545        | .7448        | .7399        | .7339        | .8171        | .8293        | .8362        | .8387        | .8423        | .5599        | .5751        | .5819        | .5832        | .5879        | 1.029                    | 1.027 | 1.024 | 1.021 | 1.02 |
| DDFM                                               | .7565        | .7510        | .7426        | .7349        | .7305        | .8275        | .8314        | .8367        | .8412        | .8436        | .5704        | .5742        | .5809        | .5873        | .5910        | 1.041                    | 1.038 | 1.036 | 1.034 | 1.03 |
| Oracle ( <i>not causal</i> )                       | .7575        | .7496        | .7393        | .7301        | .7253        | .8256        | .8311        | .8382        | .8438        | .8465        | .5715        | .5775        | .5847        | .5915        | .5953        | 1.032                    | 1.031 | 1.032 | 1.031 | 1.03 |

monotone rescaling of the raw metric and cannot reorder methods: measured across all three candidate anchors, at all five rungs, the Spearman correlation between RI and the raw metric is exactly 1.000. Whatever RI adds, it is not ranking information — its entire content is the normalisation.

That normalisation, however, is not stable. Table 4 gives the denominator for the three anchors the literature and this paper make available. TWICE’s Eq. (23) divides by the VANILLA-to-Oracle gap, but VANILLA duplicates (Sect. 5.2) and consequently sits only 0.0031–0.0053 AUC below the Oracle; dividing by a gap that small amplifies every difference four- to five-fold and throws RI outside  $[0, 100]$  in 34 of 50 cells, reaching –518%. The two admissible anchors have gaps between 2.7 and 6.2 times larger, and flatter across the ladder — their gap varies by 17% (WAIT) and 33% (FRESH) of its mean, against 59% for VANILLA — and their RI stays in range in 38 and 43 of 50 cells respectively. Since the same method can be assigned an RI of –346% or +16% at the same rung depending only on which arm is called “Vanilla”, RI is not comparable across papers unless the anchor is stated; it is a property of the denominator at least as much as of the method.

We therefore report both admissible anchors in Table 3, and they measure different things.  $RI^{\text{FRESH}}$  holds  $\Delta = o$  fixed and varies only the target, so its denominator is exactly the span of the hardness axis: it asks what fraction of the available *label-truth* gain a target captures.  $RI^{\text{WAIT}}$  compares against emitting at  $\Delta = v$  with the true label, the admissible production status quo, and so asks what fraction of the *freshness* gain is recovered. Read either way the conclusions of this section are unchanged, and two of them become legible at a glance: SOFT falls from 119% to 21% across the ladder, which is the first crossover in a single number, and the hybrids exceed 100% at S0 and

**Table 3.** Relative improvement under both admissible anchors,  $v = 7$  d; RI is Eq. (5), and 0 marks the anchor itself.  $\text{RI}^{\text{WAIT}}$  is the recovered fraction of the freshness gap,  $\text{RI}^{\text{FRESH}}$  of the label-truth gap. Values above 100 are the Oracle inversion of Sect. 5.4, not defects; FRESH’s negative  $\text{RI}^{\text{WAIT}}$  says waiting beats a censored label. The VANILLA-anchored form of Eq. (23) is reported in Table 4

| method                                             | $\text{RI}^{\text{WAIT}}_{\text{AUC}}$ |            |           |           |           | $\text{RI}^{\text{WAIT}}_{\text{NE}}$ |            |           |           |           | $\text{RI}^{\text{FRESH}}_{\text{AUC}}$ |            |           |           |           | $\text{RI}^{\text{FRESH}}_{\text{NE}}$ |            |           |           |           |
|----------------------------------------------------|----------------------------------------|------------|-----------|-----------|-----------|---------------------------------------|------------|-----------|-----------|-----------|-----------------------------------------|------------|-----------|-----------|-----------|----------------------------------------|------------|-----------|-----------|-----------|
|                                                    | S0                                     | S1         | S2        | S3        | S4        | S0                                    | S1         | S2        | S3        | S4        | S0                                      | S1         | S2        | S3        | S4        | S0                                     | S1         | S2        | S3        | S4        |
| <i>hard: target is a realised label</i>            |                                        |            |           |           |           |                                       |            |           |           |           |                                         |            |           |           |           |                                        |            |           |           |           |
| FRESH                                              | -20                                    | -19        | -18       | -39       | -72       | -360                                  | -334       | -315      | -349      | -376      | 0                                       | 0          | 0         | 0         | 0         | 0                                      | 0          | 0         | 0         | 0         |
| REWEIGHT                                           | 74                                     | 73         | 72        | 67        | 52        | 71                                    | 72         | 70        | 66        | 54        | 78                                      | 77         | 76        | 77        | 72        | 94                                     | 93         | 93        | 92        | 90        |
| WAIT                                               | 0                                      | 0          | 0         | 0         | 0         | 0                                     | 0          | 0         | 0         | 0         | 16                                      | 15         | 15        | 28        | 42        | 78                                     | 77         | 76        | 78        | 79        |
| <i>hybrid: realised label + learned correction</i> |                                        |            |           |           |           |                                       |            |           |           |           |                                         |            |           |           |           |                                        |            |           |           |           |
| TWICE                                              | 9                                      | 40         | 28        | 12        | 4         | -4                                    | 15         | 13        | -1        | -11       | 24                                      | 49         | 39        | 37        | 44        | 77                                     | 80         | 79        | 78        | 77        |
| CORRECT                                            | 116                                    | 109        | 96        | 84        | 78        | <b>110</b>                            | 104        | <b>93</b> | 83        | 78        | 113                                     | 108        | 96        | 89        | 87        | <b>102</b>                             | 101        | <b>98</b> | 96        | 95        |
| DISTIL                                             | 111                                    | <b>111</b> | <b>97</b> | <b>86</b> | <b>80</b> | 105                                   | <b>105</b> | 93        | <b>84</b> | <b>78</b> | 109                                     | <b>110</b> | <b>97</b> | <b>90</b> | <b>88</b> | 101                                    | <b>101</b> | 98        | <b>96</b> | <b>95</b> |
| <i>soft: model estimate alone</i>                  |                                        |            |           |           |           |                                       |            |           |           |           |                                         |            |           |           |           |                                        |            |           |           |           |
| SOFT                                               | <b>119</b>                             | 87         | 71        | 38        | 21        | 101                                   | 74         | 62        | 23        | 10        | <b>116</b>                              | 89         | 76        | 56        | 54        | 100                                    | 94         | 91        | 83        | 81        |
| <i>inadmissible: &gt; 1 record per click</i>       |                                        |            |           |           |           |                                       |            |           |           |           |                                         |            |           |           |           |                                        |            |           |           |           |
| VANILLA                                            | 77                                     | 80         | 79        | 77        | 63        | 54                                    | 58         | 57        | 54        | 40        | 81                                      | 83         | 83        | 84        | 79        | 90                                     | 90         | 90        | 90        | 87        |
| ES-DFM                                             | 39                                     | 85         | 85        | 64        | 69        | 44                                    | 78         | 78        | 62        | 66        | 49                                      | 88         | 88        | 74        | 82        | 88                                     | 95         | 95        | 92        | 93        |
| DEFUSE                                             | 43                                     | 89         | 88        | 66        | 71        | 47                                    | 82         | 81        | 65        | 68        | 52                                      | 90         | 90        | 76        | 83        | 89                                     | 96         | 95        | 92        | 93        |
| DDFM                                               | 113                                    | 102        | 92        | 83        | 80        | 104                                   | 95         | 89        | 83        | 81        | 110                                     | 102        | 93        | 88        | 88        | 101                                    | 99         | 97        | 96        | 96        |
| Oracle ( <i>not causal</i> )                       | -                                      | -          | -         | -         | -         | -                                     | -          | -         | -         | -         | -                                       | -          | -         | -         | -         | -                                      | -          | -         | -         | -         |

**Table 4.** Why the RI anchor matters. The denominator of Eq. (5) for the three candidate anchors, and the resulting spread over all arms and rungs. VANILLA duplicates, which puts it within 0.0053 AUC of the Oracle and makes the quotient unstable; the two admissible anchors are 2.7 to 6.2 times further from it and flatter across the ladder

| anchor                                     | Oracle – anchor (AUC) |       |       |       |       | RI range      | outside |
|--------------------------------------------|-----------------------|-------|-------|-------|-------|---------------|---------|
|                                            | S0                    | S1    | S2    | S3    | S4    |               |         |
| VANILLA, duplicating — <i>inadmissible</i> | .0034                 | .0031 | .0035 | .0034 | .0053 | $[-518, 186]$ | 34/50   |
| FRESH, $\Delta=o$ , censored label         | .0180                 | .0184 | .0200 | .0211 | .0247 | $[15, 116]$   | 7/50    |
| WAIT, $\Delta=v$ , true label              | .0151                 | .0155 | .0170 | .0152 | .0144 | $[-72, 119]$  | 12/50   |

S1, which is the Oracle inversion restated. FRESH’s negative  $\text{RI}^{\text{WAIT}}$  is likewise real: waiting beats training on a censored label.

**Second window.** At  $v = 1$  d the ordering is unchanged and the inversion is stronger: at S0 the hybrid reaches 0.8352 and CORRECT 0.8334 against Oracle 0.8323.

## 6 Discussion: why the soft end is worth its cost

Table 2 makes the soft target look like a narrow win: best at one rung, clearly worse at four. Proposition 1(ii) is the reason to care about it anyway. A target that does not read the record’s realised label leaves that record with no label at all — only features and an estimator’s read-out — so R2 is satisfied with nothing to check.

This matters when one served model must predict several conversion events with different attribution windows, the normal situation in advertising: an install, a purchase

and a subscription renewal do not share a horizon. Under R1 they share a single record. A hard or hybrid target requires that record to carry a realised label *per objective*; all of them can be frozen at a common  $o$ , but their informativeness differs by orders of magnitude — within this one dataset the fresh label carries 77% of the target’s positives at  $v = 1$  d and 47% at  $v = 30$  d (Sect. 3) — so a single shared observation age is well matched to short-window objectives and nearly vestigial for long-window ones, while still binding every corrector to the  $P(Y^{(v)}=1 \mid Y^{(o)}=0)$  conditioning. A soft target removes both dependencies: one estimator per objective, each waiting its own  $v$  offline, and a served record carrying features only.

The cost is one additional training-time model, with serving cost unchanged, which is the trade the hybrid methods already make.

## 7 Limitations

One public corpus with a single conversion event, so the multi-objective consolidation of Sect. 6 is a consequence of Proposition 1, not an experimental result; and our measurements place the soft target ahead only below roughly  $10^5$  served parameters, which is a real restriction even though it is the regime a latency-bound ranker occupies. The constraint is one production contract among several: systems that do support replay may use the methods we rule out, and for them our audit is a cost accounting rather than a prohibition. Whether R2 binds auxiliary models is a reading we adopt explicitly (Sect. 5.2), not a fact about the constraint. The ladder varies table rows at fixed embedding width. We report  $v = 1$  and 7 d but not 30 d: on a 60-day corpus a 30-day window leaves the estimator with no matured rows during model selection, so that comparison would measure the corpus’s span-to-window ratio rather than the effect of a long window. All results are offline.

## 8 Conclusion

Delayed-feedback methods are usually compared on accuracy at one model size. We added a systems constraint that many production pipelines already impose — one training record per click, label frozen at a fixed wait — and found that nine of fourteen published methods cannot satisfy it, including the baseline the field normalises against. The constraint is clarifying rather than merely restrictive: with one record carrying one target, the remaining design freedom collapses to how much of that target is a realised label, and the best choice along that axis is set by the served model’s capacity. Below roughly  $10^5$  parameters a corrected or wholly estimated target beats the true label, and at the smallest size a target that reads no label at all is the best admissible arm; above that the ordering reverses. Which hybrid estimator supplies the correction matters far less — 0.0003 AUC — than the tuning of the baselines it is compared against, which moves 0.0022. Evaluating at a single capacity, as this literature does, hides all of this.

## A Experimental specification

### A.1 The two admissibility rules

Everything in this study is governed by two rules. They are stated first because they decide which methods may appear at all, and because one of them has a clause that is easy to get wrong.

**Rule 1 (hard labels).** A training record emitted after a wait  $\Delta$  carries the label

$$y_i = 1[\text{conv\_ts}_i \leq \text{click\_ts}_i + \Delta], \quad (6)$$

and that value is frozen. A conversion arriving after  $\text{click\_ts}_i + \Delta$  must not flip it, must not add a second supervision signal, and must not be recovered later.

**Rule 1b (teacher soft labels).** A soft target is exempt from (6) provided the model producing it is itself trained under Rule 1. Concretely: the teacher trains only on *matured* records, whose hard labels obey (6) at  $\Delta = v$ ; it is then *evaluated* on the *freshes* clicks, those released at  $\Delta = o$ , and the student trains on that read-out. This is admissible because the student’s record carries no hard label drawn from its own future — only a deterministic function of its features, computed by parameters frozen at emission.

**Rule 2 (single visit).** Each click contributes *exactly one* training record to the served model, ever.

*The causality obligation Rule 1b creates.* Rule 1b is only sound if the teacher is causal at the moment it scores. At cutoff  $T_n$  our teacher has trained on clicks satisfying  $C_i + v \leq T_n$  and on nothing else, so every parameter it holds derives from outcomes that were knowable at  $T_n$ . Scoring a click released at  $T_n$  therefore uses no future information. This is not assumed: it is enforced in code and checked by gate G1 of `tr_verify_distill.py`, which fails the run if any teacher update reads a row whose window closes after the current cutoff.

*What Rule 1b does not permit.* The teacher may not be trained on the fresh stream, and the student may not consume the teacher’s read-out for a click whose window has since closed. The non-compliant reference WAIT-hybrid, which trains the teacher on matured and fresh records, measures 2.08 visits per click and is reported only to bound what the rules cost.

### A.2 Per-method compliance

Table 5 classifies every method against both rules. Record counts are *measured* over a full 720-hour replay, not asserted from the description.

### A.3 Common substrate

*Corpus and split.* Criteo-DFD [1]: 15,609,913 clicks with conversion timestamps, of which 3,538,028 ever convert at any horizon and the longest observed delay is 30.1 days. Pre-train on days  $[0, 30)$ , then stream  $[30, 60)$  with 719 hourly cut-offs. At each cut-off a model consumes the records released in that hour and then predicts every click of the next hour exactly once, scored retrospectively against  $Y^{(v)}$ .

**Table 5.** Every method against both rules. <sup>1b</sup> marks a soft target admissible under Rule 1b: the producing model is itself trained under Rule 1. Records per click are measured over a full 720-hour replay

| method   | target $\tilde{y}$                                         | Rule 1          | Rule 2 | rec./click |
|----------|------------------------------------------------------------|-----------------|--------|------------|
| FRESH    | $\tilde{y} = Y^{(o)}$                                      | ✓               | ✓      | 1.00       |
| REWEIGHT | $\tilde{y} = Y^{(o)}, \text{loss} \times w_{\text{rw}}(x)$ | ✓               | ✓      | 1.00       |
| WAIT     | $\tilde{y} = Y^{(v)}$ at $\Delta=v$                        | ✓               | ✓      | 1.00       |
| TWICE    | $\tilde{y} = Y^{(o)} + \text{delay head}$                  | ✓               | ✓      | 1.00       |
| CORRECT  | $\tilde{y} = Y^{(o)} + (1 - Y^{(o)}) w(x)$                 | ✓               | ✓      | –          |
| DISTIL   | $\tilde{y} = Y^{(o)} + (1 - Y^{(o)}) w^T(x)$               | ✓ <sup>1b</sup> | ✓      | –          |
| SOFT     | $\tilde{y} = \hat{p}_v(x)$                                 | ✓ <sup>1b</sup> | ✓      | –          |
| VANILLA  | $Y^{(o)}$ plus a late positive                             | ×               | ×      | 1.08       |
| ES-DFM   | duplicated delayed positive                                | ×               | ×      | 1.08       |
| DEFUSE   | duplicated stream + correction                             | ×               | ×      | 1.08       |
| DDFM     | both streams together                                      | ×               | ×      | 1.89       |
| ORACLE   | $\tilde{y} = Y^{(v)}$ at $\Delta=o$                        | ×               | ✓      | 1.00       |

*Seeds do not change the data.* The split, the cut-offs and the per-hour cohorts are deterministic; a seed varies only parameter initialisation and minibatch order. This was verified rather than assumed: across all five seeds at a fixed rung the evaluation set is identical ( $n_{\text{eval}} = 8,196,131$ , base rate 0.136139, 719 cohorts, all bit-identical). There is therefore one data specification, not five.

*Label statistics.* Table 6 gives the label composition at both windows. The arithmetic is asserted in code: positives plus negatives equal  $N$ , the fresh positives are a strict subset of the served positives, and the observed rate equals  $\text{base} \times F(o)$  to machine precision.

Two facts drive much of the paper. First,  $Y^{(o)}$  does not depend on  $v$ , so the fresh label is *identical* at both windows (1,649,879 positives); only the target it is asked to predict moves. Second, censoring deepens with the window —  $F(o)$  falls from 0.769 at  $v = 1$  d to 0.574 at  $v = 7$  d — so the fresh label is a good proxy at one day and a poor one at seven.

*Record timing.* Figure 2 shows, for one click, when its features are known, when each candidate label becomes knowable, and when it is scored.

#### A.4 The A0–A4 capacity ladder

The previous S0–S4 ladder varied only hash rows against a *fixed* trunk of 143,293 parameters, so total capacity spanned just  $66\times$  and was trunk-dominated below the top rung; it was a table-resolution axis described as a capacity axis. A0–A4 pairs a rung of the compute-efficient frontier with a row count and lets the embedding width rise monotonically, giving  $\sim 10\times$  per step over  $10,346\times$ . A4 is byte-identical to S4, so the two ladders share a top rung.

**Table 6.** Label composition of the replay at both windows.  $Y^{(o)}$  does not depend on  $v$ , so the fresh label is identical at both; only the target it must predict moves

| quantity                                               | $v = 1 \text{ d}$ | $v = 7 \text{ d}$ |
|--------------------------------------------------------|-------------------|-------------------|
| <i>clicks in the replay</i> $[0, 60) d$                |                   |                   |
| clicks $N$                                             | 15,609,913        | 15,609,913        |
| ever converts (any horizon)                            | 3,538,028         | 3,538,028         |
| <i>served target</i> $Y^{(v)} = 1[D \leq v]$           |                   |                   |
| positives                                              | 2,145,812         | 2,875,121         |
| negatives                                              | 13,464,101        | 12,734,792        |
| positive rate                                          | 0.1375            | 0.1842            |
| <i>fresh label</i> $Y^{(o)} = 1[D \leq o]$ , $o = 1 h$ |                   |                   |
| positives                                              | 1,649,879         | 1,649,879         |
| negatives                                              | 13,960,034        | 13,960,034        |
| positive rate                                          | 0.1057            | 0.1057            |
| censored positives ( $Y^{(v)}=1, Y^{(o)}=0$ )          | 495,933           | 1,225,242         |
| $F(o) = P(D \leq o \mid D \leq v)$                     | 0.7689            | 0.5738            |
| <i>delay of within-window converters (hours)</i>       |                   |                   |
| median                                                 | 0.17              | 0.44              |
| p90                                                    | 8.56              | 89.67             |
| p99                                                    | 23.17             | 160.67            |

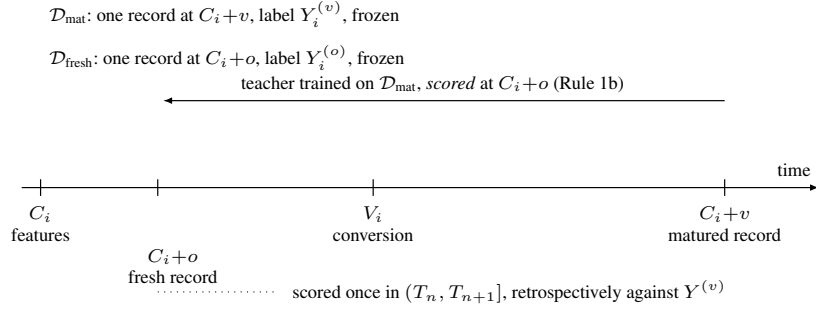
*Why the width rises.* Two earlier designs were rejected on measurement. Letting each frontier node keep its own width put A3 at  $\text{emb} = 48$  with 2048 rows, which scored worse than A2 at  $\text{emb} = 8$  with 1280 rows despite ten times the parameters: on this corpus resolution carries the signal and width does not. Pinning the width at 8 everywhere fixed that but removed the Oracle inversion entirely ( $+0.0166 \rightarrow -0.0004$  at the bottom rung), because the inversion needs genuine representational starvation and  $\text{emb} = 8$  is already wide enough at 36 rows. The adopted ladder raises width monotonically,  $2 \rightarrow 4 \rightarrow 8$ , keeping the starved bottom without the mid-ladder misallocation.

### A.5 Experiment matrix

Learning rates are selected *per arm, per rung, per window* from  $\{10^{-6}, 2.5 \times 10^{-6}, 10^{-5}, 5 \times 10^{-5}, 2 \times 10^{-4}, 5 \times 10^{-4}, 10^{-3}\}$  on a held-out chronological slice of the pre-training period, and auxiliary learning rates are swept per method for the reason established in Sect. 5.3. Curves are recorded at seed 0; all reported statistics use five seeds and paired tests over matched seeds.

## B Per-method specification

Each section states the method’s target, its compliance, the auxiliary it adds to the shared A0–A4 trunk, and the composition of the data it actually consumes. Record counts are measured; label counts are computed from the corpus.



**Fig. 2.** Timing for a single click. Features are known at  $C_i$ . The fresh pipeline writes one record at  $C_i + o$  carrying  $Y^{(o)}$ ; the matured pipeline writes one at  $C_i + v$  carrying  $Y^{(v)}$ . Rule 2 permits the served model only one of them. Under Rule 1b the teacher learns from the matured pipeline but is *evaluated* at the fresh release, so the student’s record carries a soft target and no future hard label

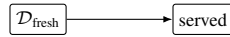
**Table 7.** The A0–A4 ladder, introspected from the instantiated models. The trunk grows with the tables, and embedding width rises monotonically  $2 \rightarrow 4 \rightarrow 8$

| rung | frontier node | emb. dim | table rows | tables    | trunk   | total     |
|------|---------------|----------|------------|-----------|---------|-----------|
| A0   | n0            | 2        | 36         | 648       | 278     | 926       |
| A1   | n2            | 4        | 256        | 9,216     | 594     | 9,810     |
| A2   | n4            | 8        | 1,280      | 92,160    | 2,799   | 94,959    |
| A3   | n6            | 8        | 16,384     | 1,179,648 | 4,791   | 1,184,439 |
| A4   | –             | 8        | 131,072    | 9,437,184 | 143,293 | 9,580,477 |

## B.1 FRESH

*Definition.* The uncorrected floor. One record per click at  $\Delta = o$  carrying the observed label, trained with binary cross-entropy. It obeys both rules trivially and is the reference against which the value of *label truth* is measured, since it differs from the Oracle only in what the record’s label says.

$$\tilde{y}_i = Y_i^{(o)} \quad (7)$$



*Compliance.* Measured at 1.00 training records per click (maximum 1) over the 720-hour replay. Single-visit; admissible under Rule 2.

*Architecture.* The served model is the A0–A4 ladder of Table 7, unchanged. No auxiliary model.

*Training data.*

**Table 8.** What the production run executes. Seeds replicate initialisation and shuffling only; the data is identical across them

| axis                                                    | levels                                              | count |
|---------------------------------------------------------|-----------------------------------------------------|-------|
| method                                                  | the 12 arms of Table 5                              | 12    |
| capacity                                                | A0, A1, A2, A3, A4                                  | 5     |
| window                                                  | $v = 1 \text{ d}, v = 7 \text{ d}$                  | 2     |
| seed                                                    | $0 \dots 4$ (init and shuffle only; data identical) | 5     |
| streaming runs                                          |                                                     | 600   |
| learning-rate selections (7 rates, per arm/rung/window) |                                                     | 840   |
| NE curves recorded (seed 0)                             |                                                     | 120   |

|                              | $v = 1 \text{ d}$ | $v = 7 \text{ d}$ |
|------------------------------|-------------------|-------------------|
| fresh records consumed       | 8,197,914         | 8,197,914         |
| hard positives ( $Y^{(o)}$ ) | 852,964           | 852,964           |

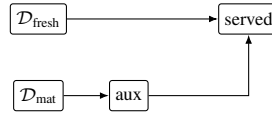
*Experiment setup.* Five rungs A0–A4  $\times$  two windows  $\times$  five seeds, learning rate selected per arm, per rung and per window from the grid of Sect. A.5. NE against streaming hour is recorded at seed 0.

*Results.* [pending the production run]

## B.2 REWEIGHT

*Definition.* FSIW [14] leaves the target alone and corrects the feedback shift in the *loss*, weighting each example by a ratio estimated from two classifiers fitted on matured data. The record and its label are untouched, so it sits at the hard end of the axis by a different route from FRESH.

$$\tilde{y}_i = Y_i^{(o)}, \quad \ell_i \mapsto w_{\text{iw}}(x_i) \ell_i \quad (8)$$



*Compliance.* Measured at 1.00 training records per click (maximum 1) over the 720-hour replay. Single-visit; admissible under Rule 2.

*Architecture.* The served model is the A0–A4 ladder of Table 7, unchanged. Auxiliary: two importance-weight classifiers, AUXHEAD, which does *not* scale with the rung, so the capacity axis measures the served model only.

*Training data.*

|                                   | $v = 1 \text{ d}$ | $v = 7 \text{ d}$ |
|-----------------------------------|-------------------|-------------------|
| fresh records consumed            | 8,197,914         | 8,197,914         |
| hard positives ( $Y^{(o)}$ )      | 852,964           | 852,964           |
| matured records for the auxiliary | 8,142,195         | 8,001,044         |

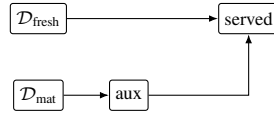
*Experiment setup.* Five rungs A0–A4  $\times$  two windows  $\times$  five seeds, learning rate selected per arm, per rung and per window from the grid of Sect. A.5. NE against streaming hour is recorded at seed 0.

*Results.* [pending the production run]

### B.3 WAIT

*Definition.* Emit nothing until the window closes, then write one record carrying the true label. The label is correct and Rule 1 is satisfied at  $\Delta = v$ , but the freshest data the model has ever seen is  $v$  old. This is the admissible production status quo and the anchor for  $\text{RI}^{\text{WAIT}}$ .

$$\tilde{y}_i = Y_i^{(v)} \text{ emitted at } \Delta = v \quad (9)$$



*Compliance.* Measured at 1.00 training records per click (maximum 1) over the 720-hour replay. Single-visit; admissible under Rule 2.

*Architecture.* The served model is the A0–A4 ladder of Table 7, unchanged. No auxiliary model.

*Training data.*

|                         | $v = 1 \text{ d}$ | $v = 7 \text{ d}$ |
|-------------------------|-------------------|-------------------|
| records consumed        | 8,142,195         | 8,001,044         |
| positives ( $Y^{(v)}$ ) | 1,107,281         | 1,477,736         |

*Experiment setup.* Five rungs A0–A4  $\times$  two windows  $\times$  five seeds, learning rate selected per arm, per rung and per window from the grid of Sect. A.5. NE against streaming hour is recorded at seed 0.

*Results.* [pending the production run]

## B.4 TWICE

*Definition.* TWICE [8] keeps its CVR tower on a single fresh record and routes conversion-clock arrivals to a separate delay head, so the served tower stays single-visit. The delay head reads the conversion clock, which is why its R2 status is boundary rather than clean.

$$\tilde{y}_i = Y_i^{(o)} \text{ plus a separate delay head on the conversion clock} \quad (10)$$



*Compliance.* Measured at 1.00 training records per click (maximum 1) over the 720-hour replay. Single-visit; admissible under Rule 2.

*Architecture.* The served model is the A0–A4 ladder of Table 7, unchanged. Auxiliary: delay head, ReLU MLP [64, 32], no shared parameters, which does *not* scale with the rung, so the capacity axis measures the served model only.

*Training data.*

|                              | $v = 1 \text{ d}$ | $v = 7 \text{ d}$ |
|------------------------------|-------------------|-------------------|
| fresh records consumed       | 8,197,914         | 8,197,914         |
| hard positives ( $Y^{(o)}$ ) | 852,964           | 852,964           |

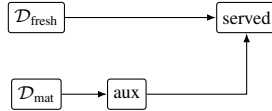
*Experiment setup.* Five rungs A0–A4  $\times$  two windows  $\times$  five seeds, learning rate selected per arm, per rung and per window from the grid of Sect. A.5. NE against streaming hour is recorded at seed 0.

*Results.* [pending the production run]

## B.5 CORRECT

*Definition.* ULC [11] keeps the realised label and softens the observed negatives by the probability they convert later,  $w(x) = P(Y^{(v)}=1 \mid Y^{(o)}=0, x)$ , estimated online on matured data. An observed positive stays exactly 1; only negatives move. This is the hybrid point of the axis and the comparator that matters for DISTIL.

$$\tilde{y}_i = Y_i^{(o)} + (1 - Y_i^{(o)}) w(x_i) \quad (11)$$



*Compliance.* Measured at 1.00 training records per click (maximum 1) over the 720-hour replay. Single-visit; admissible under Rule 2.

*Architecture.* The served model is the A0–A4 ladder of Table 7, unchanged. Auxiliary: correction head  $w$ , AUXHEAD, 9,444,801 parameters, which does *not* scale with the rung, so the capacity axis measures the served model only.

*Training data.*

|                                   | $v = 1 \text{ d}$ | $v = 7 \text{ d}$ |
|-----------------------------------|-------------------|-------------------|
| fresh records consumed            | 8,197,914         | 8,197,914         |
| hard positives ( $Y^{(o)}$ )      | 852,964           | 852,964           |
| matured records for the auxiliary | 8,142,195         | 8,001,044         |

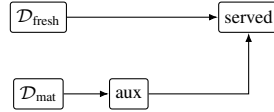
*Experiment setup.* Five rungs A0–A4  $\times$  two windows  $\times$  five seeds, learning rate selected per arm, per rung and per window from the grid of Sect. A.5. NE against streaming hour is recorded at seed 0.

*Results.* [pending the production run]

## B.6 DISTIL

*Definition.* Identical in form to CORRECT; the only difference is where  $w$  comes from. Here it is the second read-out of a teacher trained offline on the matured pipeline and *evaluated at the fresh release* under Rule 1b. The two correctors hold the same parameter count to within 0.13%, so a comparison between them isolates the estimator and nothing else.

$$\tilde{y}_i = Y_i^{(o)} + (1 - Y_i^{(o)}) w^T(x_i) \quad (12)$$



*Compliance.* Measured at 1.00 training records per click (maximum 1) over the 720-hour replay. Single-visit; admissible under Rule 2.

*Architecture.* The served model is the A0–A4 ladder of Table 7, unchanged. Auxiliary: offline teacher, two read-outs on one trunk, 9,456,897 parameters, which does *not* scale with the rung, so the capacity axis measures the served model only.

*Training data.*

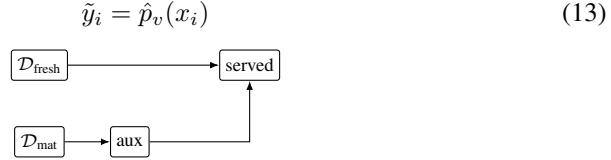
|                                   | $v = 1 \text{ d}$ | $v = 7 \text{ d}$ |
|-----------------------------------|-------------------|-------------------|
| fresh records consumed            | 8,197,914         | 8,197,914         |
| hard positives ( $Y^{(o)}$ )      | 852,964           | 852,964           |
| matured records for the auxiliary | 8,142,195         | 8,001,044         |

*Experiment setup.* Five rungs A0–A4  $\times$  two windows  $\times$  five seeds, learning rate selected per arm, per rung and per window from the grid of Sect. A.5. NE against streaming hour is recorded at seed 0.

*Results.* [pending the production run]

## B.7 SOFT

*Definition.* The extreme soft end: the target is the teacher’s estimate alone and the record carries *no realised label at all*. By Proposition 1(ii) Rule 1 is satisfied vacuously — there is no hard label for a late conversion to revise. This is the arm that makes the served record label-free, and the one whose behaviour changes most across the ladder.



*Compliance.* Measured at 1.00 training records per click (maximum 1) over the 720-hour replay. Single-visit; admissible under Rule 2.

*Architecture.* The served model is the A0–A4 ladder of Table 7, unchanged. Auxiliary: same teacher; the  $p_v$  read-out is consumed instead of  $w$ , which does *not* scale with the rung, so the capacity axis measures the served model only.

*Training data.*

|                                   | $v = 1 \text{ d}$ | $v = 7 \text{ d}$ |
|-----------------------------------|-------------------|-------------------|
| fresh records consumed            | 8,197,914         | 8,197,914         |
| hard positives ( $Y^{(o)}$ )      | 852,964           | 852,964           |
| matured records for the auxiliary | 8,142,195         | 8,001,044         |

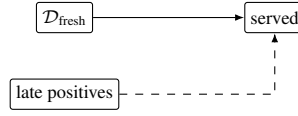
*Experiment setup.* Five rungs A0–A4  $\times$  two windows  $\times$  five seeds, learning rate selected per arm, per rung and per window from the grid of Sect. A.5. NE against streaming hour is recorded at seed 0.

*Results.* [pending the production run]

## B.8 VANILLA

*Definition.* The field’s reference baseline. Described as training each click once at age  $o$ , but reproducing any published VANILLA row requires also emitting the late conversion as a positive, which is a second visit. It is reported for continuity with the literature and is *inadmissible* here.

$$Y_i^{(o)} \text{ plus a second, positive record when the conversion lands} \quad (14)$$



*Compliance.* Measured at 1.08 training records per click (maximum 2) over the 720-hour replay. Exceeds one record per click; inadmissible under Rule 2.

*Architecture.* The served model is the A0–A4 ladder of Table 7, unchanged. No auxiliary model.

*Training data.*

|                                | $v = 1 \text{ d}$ | $v = 7 \text{ d}$ |
|--------------------------------|-------------------|-------------------|
| fresh records consumed         | 8,197,914         | 8,197,914         |
| hard positives ( $Y^{(o)}$ )   | 852,964           | 852,964           |
| extra records from duplication | 262,974           | 664,150           |

*Experiment setup.* Five rungs A0–A4  $\times$  two windows  $\times$  five seeds, learning rate selected per arm, per rung and per window from the grid of Sect. A.5. NE against streaming hour is recorded at seed 0.

*Results.* [pending the production run]

## B.9 ES-DFM

*Definition.* ES-DFM [13] samples an elapsed-time cut-off and duplicates the delayed positive into the stream, then corrects with importance weights. Duplication is the mechanism, not an implementation detail.

elapsed-time cut-off; the delayed positive is duplicated (15)



*Compliance.* Measured at 1.08 training records per click (maximum 2) over the 720-hour replay. Exceeds one record per click; inadmissible under Rule 2.

*Architecture.* The served model is the A0–A4 ladder of Table 7, unchanged. Auxiliary: delayed-positive rate, AUXHEAD, which does *not* scale with the rung, so the capacity axis measures the served model only.

*Training data.*

|                                | $v = 1 \text{ d}$ | $v = 7 \text{ d}$ |
|--------------------------------|-------------------|-------------------|
| fresh records consumed         | 8,197,914         | 8,197,914         |
| hard positives ( $Y^{(o)}$ )   | 852,964           | 852,964           |
| extra records from duplication | 262,974           | 664,150           |

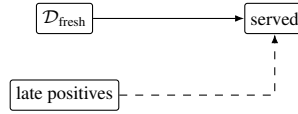
*Experiment setup.* Five rungs A0–A4  $\times$  two windows  $\times$  five seeds, learning rate selected per arm, per rung and per window from the grid of Sect. A.5. NE against streaming hour is recorded at seed 0.

*Results.* [pending the production run]

## B.10 DEFUSE

*Definition.* DEFUSE [2] separates immediate from delayed positives and corrects the duplicated stream with an unbiased estimator.

duplicated stream with a label correction on top (16)



*Compliance.* Measured at 1.08 training records per click (maximum 2) over the 720-hour replay. Exceeds one record per click; inadmissible under Rule 2.

*Architecture.* The served model is the A0–A4 ladder of Table 7, unchanged. Auxiliary: four-class mass estimator, AUXHEAD, which does *not* scale with the rung, so the capacity axis measures the served model only.

*Training data.*

|                                | $v = 1 \text{ d}$ | $v = 7 \text{ d}$ |
|--------------------------------|-------------------|-------------------|
| fresh records consumed         | 8,197,914         | 8,197,914         |
| hard positives ( $Y^{(o)}$ )   | 852,964           | 852,964           |
| extra records from duplication | 262,974           | 664,150           |

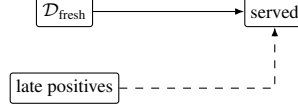
*Experiment setup.* Five rungs A0–A4  $\times$  two windows  $\times$  five seeds, learning rate selected per arm, per rung and per window from the grid of Sect. A.5. NE against streaming hour is recorded at seed 0.

*Results.* [pending the production run]

### B.11 DDFM

*Definition.* DDFM [3] runs both streams simultaneously, which is why its record count is the highest of the group at 1.89 per click. It is the strongest inadmissible method and therefore bounds what the rules cost.

fresh and delayed-positive streams trained together (17)



*Compliance.* Measured at 1.89 training records per click (maximum 2) over the 720-hour replay. Exceeds one record per click; inadmissible under Rule 2.

*Architecture.* The served model is the A0–A4 ladder of Table 7, unchanged. Auxiliary: dual heads, AUXHEAD, which does *not* scale with the rung, so the capacity axis measures the served model only.

*Training data.*

|                                | $v = 1$ d | $v = 7$ d |
|--------------------------------|-----------|-----------|
| fresh records consumed         | 8,197,914 | 8,197,914 |
| hard positives ( $Y^{(o)}$ )   | 852,964   | 852,964   |
| extra records from duplication | 262,974   | 664,150   |

*Experiment setup.* Five rungs A0–A4  $\times$  two windows  $\times$  five seeds, learning rate selected per arm, per rung and per window from the grid of Sect. A.5. NE against streaming hour is recorded at seed 0.

*Results.* [pending the production run]

### B.12 ORACLE

*Definition.* Not causal and not deployable: the fresh record is labelled with an outcome that had not arrived when it was written. It exists to bound the recoverable gap. Sect. 5.4 shows it is *not* an upper bound at small capacity, which is the paper’s most surprising result.

$$\tilde{y}_i = Y_i^{(v)} \text{ emitted at } \Delta = o \quad (18)$$



*Compliance.* Measured at 1.00 training records per click (maximum 1) over the 720-hour replay. Single-visit; admissible under Rule 2.

*Architecture.* The served model is the A0–A4 ladder of Table 7, unchanged. No auxiliary model.

*Training data.*

|                               | $v = 1 \text{ d}$ | $v = 7 \text{ d}$ |
|-------------------------------|-------------------|-------------------|
| fresh records consumed        | 8,197,914         | 8,197,914         |
| hard positives ( $Y^{(o)}$ )  | 852,964           | 852,964           |
| label positives ( $Y^{(v)}$ ) | 1,115,938         | 1,517,114         |

*Experiment setup.* Five rungs A0–A4  $\times$  two windows  $\times$  five seeds, learning rate selected per arm, per rung and per window from the grid of Sect. A.5. NE against streaming hour is recorded at seed 0.

*Results.* [pending the production run]

## References

1. Chapelle, O.: Modeling delayed feedback in display advertising. In: Proc. KDD. pp. 1097–1105 (2014)
2. Chen, Y., Jin, J., Zhao, H., Wang, P., Liu, G., Xu, J., Zheng, B.: Asymptotically unbiased estimation for delayed feedback modeling via label correction. In: Proc. WWW. pp. 369–379 (2022)
3. Dai, S., Zhou, Y., Xu, J., Wen, J.R.: Dually enhanced delayed feedback modeling for streaming conversion rate prediction. In: Proc. CIKM. pp. 390–399 (2023)
4. Ding, C., Wu, J., Yuan, Y., Li, C., Wang, X., Wang, D., Yang, F., Rabinovich, A.: Delayed feedback modeling with influence functions. Proc. AAAI **40**(17), 14648–14656 (2026)
5. Gu, S., Sheng, X.R., Fan, Y., Zhou, G., Zhu, X.: Real negatives matter: Continuous training with real negatives for delayed feedback modeling. In: Proc. KDD. pp. 2890–2898 (2021)
6. Ktena, S.I., Tejani, A., Theis, L., Myana, P.K., Dilipkumar, D., Huszar, F., Yoo, S., Shi, W.: Addressing delayed feedback for continuous training with neural networks in ctr prediction. In: Proc. RecSys. pp. 187–195 (2019)
7. Li, H., Pan, F., Ao, X., Yang, Z., Lu, M., Pan, J., Liu, D., Xiao, L., He, Q.: Follow the prophet: Accurate online conversion rate prediction in the face of delayed feedback. In: Proc. SIGIR. pp. 1915–1919 (2021)
8. Li, K., Wang, K., Wang, Z., Sha, T., Yan, M., Cheng, Y., Liu, X.: TWICE: Two-clock, two-window learning for long-horizon conversion prediction in online advertising (2026)
9. Liu, Q., Ao, X., Guo, Y., He, Q.: Online conversion rate prediction via multi-interval screening and synthesizing under delayed feedback. Proc. AAAI **38**(8), 8796–8804 (2024)
10. Menon, et al.: A statistical perspective on distillation. In: Proc. ICML (2021)
11. Wang, Y., Sun, P., Zhang, M., Jia, Q., Li, J., Ma, S.: Unbiased delayed feedback label correction for conversion rate prediction. In: Proc. KDD. pp. 2456–2466 (2023)
12. Xu, et al.: Privileged features distillation at Taobao recommendations (2019)
13. Yang, J.Q., Li, X., Han, S., Zhuang, T., Zhan, D.C., Zeng, X., Tong, B.: Capturing delayed feedback in conversion rate prediction via elapsed-time sampling. Proc. AAAI **35**(5), 4582–4589 (2021)
14. Yasui, S., Morishita, G., Fujita, K., Shibata, M.: A feedback shift correction in predicting conversion rates under delayed feedback. In: Proc. WWW. pp. 2740–2746 (2020)

15. Zhang, X., Li, W., Li, R., Fu, Z., Tang, T., Zhang, Z., Chen, W.Y., Noorshams, N., Jasapara, N., Ding, X., Wen, E., Feng, X.: Personalized interpolation: Achieving efficient conversion estimation with flexible optimization windows. In: Proc. CIKM (2025)